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# RoboCup@Home

Forms & Score Sheets

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# Registration Form



**Team name:** \_\_\_\_\_

**Team leader name:** \_\_\_\_\_

## Safety first!

The speed of the robots should ensure safe operation. This means that the robot is not allowed to go faster than the local speed limits. During operation in an environment with humans it may not exceed a walking pace of humans (6 kph, 3.7 mph).

At any time when operating the robot inside and outside the scenario the owners have to stop the robot immediately if there is a remote possibility of dangerous behavior towards people and/or objects. If a referee, member of the Technical or Organizational committee, an Executive or Trustee of the federation tells the team to stop the robot, there will be no discussion and the robot has to be stopped *immediately*.

If the team does not comply, the team and its members can be excluded from the ongoing competition immediately by a decision of the RoboCup@Home Technical Committee. Furthermore, the team and its members can be banned from future competitions for a period not less than a year by a decision of the RoboCup Federation Trustee Board.

|                         | Robot 1 | Robot 2 |
|-------------------------|---------|---------|
| Size of the robot       |         |         |
| Emergency button(s)     |         |         |
| Start button            |         |         |
| Collision avoidance     |         |         |
| Voice of the robot      |         |         |
| Robot speaker system    |         |         |
| Use of external devices |         |         |
| ASR Bypassing method    |         |         |
| Other safety issues     |         |         |
| Custom containers.      |         |         |

## External Devices:

I have read and fully understood the “Safety First” statement and agree to it. Furthermore, I agree to obey the rules in the official RoboCup@Home RuleBook.

\_\_\_\_\_  
Date & time

\_\_\_\_\_  
Team leader

# Score Sheet



**Test:** Poster Session

**Team name:** \_\_\_\_\_

**Referee name:** \_\_\_\_\_

## Instructions:

1. Please watch all posters carefully.
2. There will be a team member for questions about the poster.
3. After viewing the poster and asking your questions you may have, please fill in the evaluation sheet below.
4. Is not allowed to evaluate your own team.
5. You may use the space below "Remarks:" to take notes for yourself.
6. Enter your team's name ("Team name") and your name ("referee name") on top of the sheet.
7. Sign the form using either "Referee" or "Team leader" slots at the bottom.

## Evaluation sheet

| Team                                  | Overall score<br>of poster (0-10) |
|---------------------------------------|-----------------------------------|
| Chief Scientist Office                |                                   |
| EPFL@Home                             |                                   |
| eR@sers                               |                                   |
| FBOT@Home                             |                                   |
| Gentlebots                            |                                   |
| Hibikino-Musashi@Home                 |                                   |
| KIARO                                 |                                   |
| LAR@Home                              |                                   |
| LASR                                  |                                   |
| LCASTOR                               |                                   |
| LyonTech                              |                                   |
| NimbRo@Home                           |                                   |
| Pumas                                 |                                   |
| PUMAS-DSPL                            |                                   |
| RoboCanes-VISAGE                      |                                   |
| RoboFEI@Home                          |                                   |
| RoBorregos rUNSWEEP                   |                                   |
| Serious Cybernetics Corporation (SCC) |                                   |
| Sinfonia Uniandes                     |                                   |
| SKUBA                                 |                                   |
| SOBITS                                |                                   |
| SocRob@Home                           |                                   |
| SUTURO-VaB                            |                                   |
| Team of Bielefeld (ToBi)              |                                   |
| Tech United Eindhoven                 |                                   |
| Tidyboy-DSPL                          |                                   |
| Tidyboy-OPL                           |                                   |
| Tinker                                |                                   |
| TRAIL                                 |                                   |
| UT Austin Villa@Home                  |                                   |

**Remarks:**

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*Date & time*

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*Referee*

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*Team leader*

# Score Sheet



**Test:** General Purpose Service Robot

**Team name:** \_\_\_\_\_

**Referee name:** \_\_\_\_\_

| Action                                                                                                      | Score | 1 <sup>st</sup> try | 2 <sup>nd</sup> try | 3 <sup>rd</sup> try |
|-------------------------------------------------------------------------------------------------------------|-------|---------------------|---------------------|---------------------|
| <b>Main Goal</b>                                                                                            |       |                     |                     |                     |
| Understand the spoken command                                                                               | 3×80  | _____               | _____               | _____               |
| Using a custom operator                                                                                     | 3×-20 | _____               | _____               | _____               |
| Request a rephrasing                                                                                        | 6×-30 | _____               | _____               | _____               |
| Demonstrate a plan has been generated                                                                       | 3×100 | _____               | _____               | _____               |
| Solving the first command                                                                                   | 250   | _____               | _____               | _____               |
| Solving the second command                                                                                  | 250   | _____               | _____               | _____               |
| Solving the third command                                                                                   | 250   | _____               | _____               | _____               |
| Solving part of a command will apply<br>a percentage penalty according to similar<br>scores in other tests. |       | _____               | _____               | _____               |
| <b>Extra Rewards</b>                                                                                        |       |                     |                     |                     |
| Interleaved Task Bonus                                                                                      | 200   | _____               | _____               | _____               |
| <b>Penalties</b>                                                                                            |       |                     |                     |                     |
| Bypassing speech recognition                                                                                | 3×-50 | _____               | _____               | _____               |
| <b>Special Penalties &amp; Bonuses</b>                                                                      |       |                     |                     |                     |
| Not attending                                                                                               | -500  | _____               | _____               | _____               |
| Using alternative start signal                                                                              | -100  | _____               | _____               | _____               |
| Outstanding performance                                                                                     | 149   | _____               | _____               | _____               |
| <b>Recordings</b>                                                                                           |       |                     |                     |                     |
| Duration (limit: <b>7:00</b> minutes)                                                                       |       | _____               | _____               | _____               |
| Restart after (sec)                                                                                         |       | _____               | _____               | _____               |
| <hr/>                                                                                                       |       |                     |                     |                     |
| <b>Score per try</b>                                                                                        | 1490  | _____               | _____               | _____               |
| <b>Total Score</b>                                                                                          | 1639  | _____               | _____               | _____               |

**Remarks:**

\_\_\_\_\_  
Date & time

\_\_\_\_\_  
Referee

\_\_\_\_\_  
Team leader

# Score Sheet



**Test:** Human Robot Interaction Challenge

**Team name:** \_\_\_\_\_

**Referee name:** \_\_\_\_\_

| Action                                                                                  | Score  | 1 <sup>st</sup> try | 2 <sup>nd</sup> try | 3 <sup>rd</sup> try |
|-----------------------------------------------------------------------------------------|--------|---------------------|---------------------|---------------------|
| <b>Greeting</b>                                                                         |        |                     |                     |                     |
| Detect the doorbell sound as a signal that a guest has arrived                          | 2×30   | _____               | _____               | _____               |
| Open the entrance door for a guest                                                      | 2×200  | _____               | _____               | _____               |
| Look at the person talking, when receiving a guest                                      | 2×50   | _____               | _____               | _____               |
| Offer a free seat to the new guest                                                      | 2×100  | _____               | _____               | _____               |
| Look in the direction of navigation or at the navigation goal                           | 2×15   | _____               | _____               | _____               |
| Tell a visual attribute of the first guest to the second guest                          |        | _____               | _____               | _____               |
| Correct attribute                                                                       | 4×+20  | _____               | _____               | _____               |
| Incorrect attribute                                                                     | 4×-20  | _____               | _____               | _____               |
| Not asking non-essential questions to confirm or correct information                    | 4×15   | _____               | _____               | _____               |
| <b>Introduction</b>                                                                     |        |                     |                     |                     |
| Say name and favorite drink of each guest (during introduction)                         | 4×30   | _____               | _____               | _____               |
| While introducing guests, look to the correct guest while talking about the other guest | 2×50   | _____               | _____               | _____               |
| <b>Guiding</b>                                                                          |        |                     |                     |                     |
| Receive the bag from the guest via handover                                             | 50     | _____               | _____               | _____               |
| Requesting handover assistance from the guest                                           | -25    | _____               | _____               | _____               |
| Bag placed on the robot structure instead of being handed over                          | -50    | _____               | _____               | _____               |
| Following the host to the bag drop area                                                 | 200    | _____               | _____               | _____               |
| Drop the bag while following the host                                                   | -50    | _____               | _____               | _____               |
| Rediscovering the operator by natural interaction                                       | -50    | _____               | _____               | _____               |
| Asking the operator to wait                                                             | -50    | _____               | _____               | _____               |
| Guiding the robot with physical contact (take by the hand)                              | -150   | _____               | _____               | _____               |
| Drop the bag in the correct area                                                        | 50     | _____               | _____               | _____               |
| <b>Penalties</b>                                                                        |        |                     |                     |                     |
| Wrong guest information was memorized                                                   | 4×-40  | _____               | _____               | _____               |
| Alternative HRI                                                                         | 6×-20  | _____               | _____               | _____               |
| Not acknowledging people                                                                | 2×-200 | _____               | _____               | _____               |
| <b>Special Penalties &amp; Bonuses</b>                                                  |        |                     |                     |                     |
| Not attending                                                                           | -500   | _____               | _____               | _____               |
| Outstanding performance                                                                 | 145    | _____               | _____               | _____               |
| <b>Recordings</b>                                                                       |        |                     |                     |                     |
| Duration (limit: 7:00 minutes)                                                          |        | _____               | _____               | _____               |
| Restart after (sec)                                                                     |        | _____               | _____               | _____               |
| <hr/>                                                                                   |        |                     |                     |                     |
| <b>Score per try</b>                                                                    | 1450   | _____               | _____               | _____               |
| <b>Total Score</b>                                                                      | 1595   | _____               | _____               | _____               |

**Remarks:**

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*Date & time*

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*Referee*

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*Team leader*

# Score Sheet



**Test:** Pick and Place Challenge

**Team name:** \_\_\_\_\_

**Referee name:** \_\_\_\_\_



| Action                                                         | Score   | 1 <sup>st</sup> try | 2 <sup>nd</sup> try | 3 <sup>rd</sup> try |
|----------------------------------------------------------------|---------|---------------------|---------------------|---------------------|
| Navigate to the table                                          | 15      | _____               | _____               | _____               |
| Correctly recognize an object                                  | 12×10   | _____               | _____               | _____               |
| Perceive objects on a shelf and indicate the correct placement | 2×30    | _____               | _____               | _____               |
| <b>Picking</b>                                                 |         |                     |                     |                     |
| Picking up an object for transportation                        | 12×50   | _____               | _____               | _____               |
| First Pick Bonus                                               | +100    | _____               | _____               | _____               |
| From the floor                                                 | +30     | _____               | _____               | _____               |
| Cutlery                                                        | 2×+50   | _____               | _____               | _____               |
| Plate                                                          | +100    | _____               | _____               | _____               |
| Dishwasher tab                                                 | +100    | _____               | _____               | _____               |
| Common object from auxiliary table                             | 2×-20   | _____               | _____               | _____               |
| <b>Placing</b>                                                 |         |                     |                     |                     |
| Place an object in its designated location                     | 12×40   | _____               | _____               | _____               |
| Correctly in the dishwasher                                    | 3×+70   | _____               | _____               | _____               |
| Next to similar objects in the cabinet                         | 2×+20   | _____               | _____               | _____               |
| Area around breakfast items is not cleaned                     | 4×-30   | _____               | _____               | _____               |
| In the dishwasher tab slot inside the dishwasher               | +160    | _____               | _____               | _____               |
| Common object from auxiliary table                             | 2×-20   | _____               | _____               | _____               |
| <b>Extra Rewards</b>                                           |         |                     |                     |                     |
| Pull or push the dishwasher rack                               | 2×100   | _____               | _____               | _____               |
| Open or close the dishwasher door without assistance           | 2×200   | _____               | _____               | _____               |
| Open milk container without assistance                         | 400     | _____               | _____               | _____               |
| Pour cereal or milk into the bowl without assistance           | 2×200   | _____               | _____               | _____               |
| <b>Penalties</b>                                               |         |                     |                     |                     |
| Objects thrown or dropped while placing                        | 12×-40  | _____               | _____               | _____               |
| Breakfast not served in a typical meal setting                 | -50     | _____               | _____               | _____               |
| Objects dropped on the floor                                   | 12×-40  | _____               | _____               | _____               |
| Spilling cereal and milk while pouring                         | 2×-100  | _____               | _____               | _____               |
| Human assistance: object repositioned by a person              | 12×-30  | _____               | _____               | _____               |
| Human assistance: handover                                     | 24×-100 | _____               | _____               | _____               |
| Human assistance: environment changes (per item)               | -40     | _____               | _____               | _____               |
| Human assistance: opening milk container                       | -0      | _____               | _____               | _____               |
| Human assistance: moving dishwasher door or rack               | -0      | _____               | _____               | _____               |
| <b>Special Penalties &amp; Bonuses</b>                         |         |                     |                     |                     |
| Not attending                                                  | -500    | _____               | _____               | _____               |
| Using alternative start signal                                 | -100    | _____               | _____               | _____               |
| Outstanding performance                                        | 351     | _____               | _____               | _____               |
| <b>Recordings</b>                                              |         |                     |                     |                     |
| Duration (limit: <b>7:00</b> minutes)                          |         | _____               | _____               | _____               |
| Restart after (sec)                                            |         | _____               | _____               | _____               |
| <hr/>                                                          |         |                     |                     |                     |
| <b>Score per try</b>                                           | 3515    | _____               | _____               | _____               |
| <b>Total Score</b>                                             | 3866    | _____               |                     |                     |
| <b>Remarks:</b>                                                |         |                     |                     |                     |

\_\_\_\_\_  
Date & time

\_\_\_\_\_  
Referee

\_\_\_\_\_  
Team leader

# Score Sheet



Test: Restaurant

Team name: \_\_\_\_\_

Referee name: \_\_\_\_\_

| Action                                                                                   | Score  | 1 <sup>st</sup> try | 2 <sup>nd</sup> try | 3 <sup>rd</sup> try |
|------------------------------------------------------------------------------------------|--------|---------------------|---------------------|---------------------|
| <b>Regular Rewards</b>                                                                   |        |                     |                     |                     |
| Detect calling or waving customer                                                        | 2×80   | _____               | _____               | _____               |
| Reach a customer's table                                                                 | 2×80   | _____               | _____               | _____               |
| Human Assistance: Being guided to a table                                                | 2×-80  | _____               | _____               | _____               |
| Understand and confirm the order received to the customer                                | 2×160  | _____               | _____               | _____               |
| Alternative HRI                                                                          | 2×-80  | _____               | _____               | _____               |
| Not making eye-contact when taking the order                                             | 2×-60  | _____               | _____               | _____               |
| Communicate the order to the barman                                                      | 2×80   | _____               | _____               | _____               |
| Picking up the requested items from the <i>Kitchen-bar</i>                               | 4×100  | _____               | _____               | _____               |
| First Pick Bonus                                                                         | +100   | _____               | _____               | _____               |
| Human assistance: Asking the Barman to handover object to the robot                      | 4×-100 | _____               | _____               | _____               |
| Return to the customer table with the order                                              | 2×80   | _____               | _____               | _____               |
| Serve the order to the customer                                                          | 4×100  | _____               | _____               | _____               |
| First Place Bonus                                                                        | +100   | _____               | _____               | _____               |
| Human assistance: Guest needing to take the object from a tray or the robot's hand       | 4×-100 | _____               | _____               | _____               |
| <b>Extra Rewards</b>                                                                     |        |                     |                     |                     |
| Use an unattached tray to transport                                                      | 2×200  | _____               | _____               | _____               |
| <b>Penalties</b>                                                                         |        |                     |                     |                     |
| Not reaching the bar (barman has to move from behind the bar to interact with the robot) | 2×-60  | _____               | _____               | _____               |
| Human Assistance: Asking for directional confirmation                                    | 2×-30  | _____               | _____               | _____               |
| Human Assistance: Being told/pointed where a table/ <i>Kitchen-bar</i> is located        | 2×-40  | _____               | _____               | _____               |
| <b>Special Penalties &amp; Bonuses</b>                                                   |        |                     |                     |                     |
| Not attending                                                                            | -500   | _____               | _____               | _____               |
| Using alternative start signal                                                           | -100   | _____               | _____               | _____               |
| Outstanding performance                                                                  | 236    | _____               | _____               | _____               |
| <b>Recordings</b>                                                                        |        |                     |                     |                     |
| Duration (limit: 15:00 minutes)                                                          |        | _____               | _____               | _____               |
| Restart after (sec)                                                                      |        | _____               | _____               | _____               |
| <b>Score per try</b>                                                                     | 2360   | _____               | _____               | _____               |
| <b>Total Score</b>                                                                       | 2596   | _____               | _____               | _____               |

Remarks:

\_\_\_\_\_  
Date & time

\_\_\_\_\_  
Referee

\_\_\_\_\_  
Team leader

# Score Sheet



Test: Doing Laundry

Team name: \_\_\_\_\_

Referee name: \_\_\_\_\_

| Action                                                                                          | Score  | 1 <sup>st</sup> try | 2 <sup>nd</sup> try | 3 <sup>rd</sup> try |
|-------------------------------------------------------------------------------------------------|--------|---------------------|---------------------|---------------------|
| <b>Main Goal</b>                                                                                |        |                     |                     |                     |
| Navigating to the laundry area                                                                  | 15     | _____               | _____               | _____               |
| Picking up a piece of clothing from the basket                                                  | 100    | _____               | _____               | _____               |
| Picking up multiple at once                                                                     | -100   | _____               | _____               | _____               |
| Folding a piece of clothing                                                                     | 800    | _____               | _____               | _____               |
| Human assistance: Flattening/Arranging clothing before folding                                  | -200   | _____               | _____               | _____               |
| Human assistance during folding will apply a percentage penalty according to the amount of help | -800   | _____               | _____               | _____               |
| <b>Extra Rewards</b>                                                                            |        |                     |                     |                     |
| Opening the washing machine door                                                                | 300    | _____               | _____               | _____               |
| Removing one or more pieces of clothing from the washing machine                                | 300    | _____               | _____               | _____               |
| The clothing touches the floor                                                                  | -200   | _____               | _____               | _____               |
| Using the basket for transportation                                                             | 300    | _____               | _____               | _____               |
| Laundry is dropped or otherwise lost during transportation.                                     | -200   | _____               | _____               | _____               |
| Folding additional clothes (per item)                                                           | 5×400  | _____               | _____               | _____               |
| Human assistance: Flattening/Arranging clothing before folding                                  | 5×-200 | _____               | _____               | _____               |
| Human assistance during folding will apply a percentage penalty according to the amount of help | 5×-400 | _____               | _____               | _____               |
| Stacking folded clothes neatly (per item)                                                       | 6×100  | _____               | _____               | _____               |
| <b>Penalties</b>                                                                                |        |                     |                     |                     |
| Human assistance: environment changes (per item)                                                | -40    | _____               | _____               | _____               |
| <b>Special Penalties &amp; Bonuses</b>                                                          |        |                     |                     |                     |
| Not attending                                                                                   | -500   | _____               | _____               | _____               |
| Using alternative start signal                                                                  | -100   | _____               | _____               | _____               |
| Outstanding performance                                                                         | 441    | _____               | _____               | _____               |
| <b>Recordings</b>                                                                               |        |                     |                     |                     |
| Duration (limit: 7:00 minutes)                                                                  |        | _____               | _____               | _____               |
| Restart after (sec)                                                                             |        | _____               | _____               | _____               |

Score per try 4415 \_\_\_\_\_

Total Score 4856 \_\_\_\_\_

Remarks:

\_\_\_\_\_  
Date & time

\_\_\_\_\_  
Referee

\_\_\_\_\_  
Team leader

# Score Sheet



**Test:** Final Demonstration

**Team name:** \_\_\_\_\_

**Referee name:** \_\_\_\_\_

| Action                                                                                                                             | Score   | Single try |
|------------------------------------------------------------------------------------------------------------------------------------|---------|------------|
| <b><i>Main Goal (can be repeated unlimited times)</i></b>                                                                          |         |            |
| Find and clearly state an encountered problem                                                                                      | 3× 150  | _____      |
| Find repeated problem category                                                                                                     | −100    | _____      |
| Human Assistance: Asking for location of a problem                                                                                 | −150    | _____      |
| Solve a problem                                                                                                                    | 10× 650 | _____      |
| Solving repeated problem category for the 2nd time                                                                                 | −300    | _____      |
| Solving repeated problem category for the 3rd (or more) time                                                                       | −500    | _____      |
| Instructing a human to perform parts<br>of the task will apply a percentage penalty according to<br>similar scores in other tests. | −650    | _____      |
| Solving only parts of a GPSR command will apply<br>a percentage penalty according to similar<br>scores in other tests.             |         | _____      |
| Moving the laundry basket                                                                                                          | 600     | _____      |
| Opening the Door of the Apartment                                                                                                  | 600     | _____      |
| Closing the Dishwasher                                                                                                             | 300     | _____      |
| Custom:                                                                                                                            |         |            |
|                                                                                                                                    | 00      | _____      |
| Custom:                                                                                                                            |         |            |
|                                                                                                                                    | 00      | _____      |
| Custom:                                                                                                                            |         |            |
|                                                                                                                                    | 00      | _____      |
| <b><i>Penalties</i></b>                                                                                                            |         |            |
| Restart (only applies if the robot continues scoring afterwards)                                                                   | −50     | _____      |
| <b><i>Recordings</i></b>                                                                                                           |         |            |
| Duration (limit: 10:00 minutes)                                                                                                    |         | _____      |
| Restart after (sec)                                                                                                                |         | _____      |

**Total Score** 8450 \_\_\_\_\_

**Referee Information:** Note each problem (category, item, location) and mark if they are stated by the robot in remarks

**Remarks:**

\_\_\_\_\_  
Date & time

\_\_\_\_\_  
Referee

\_\_\_\_\_  
Team leader